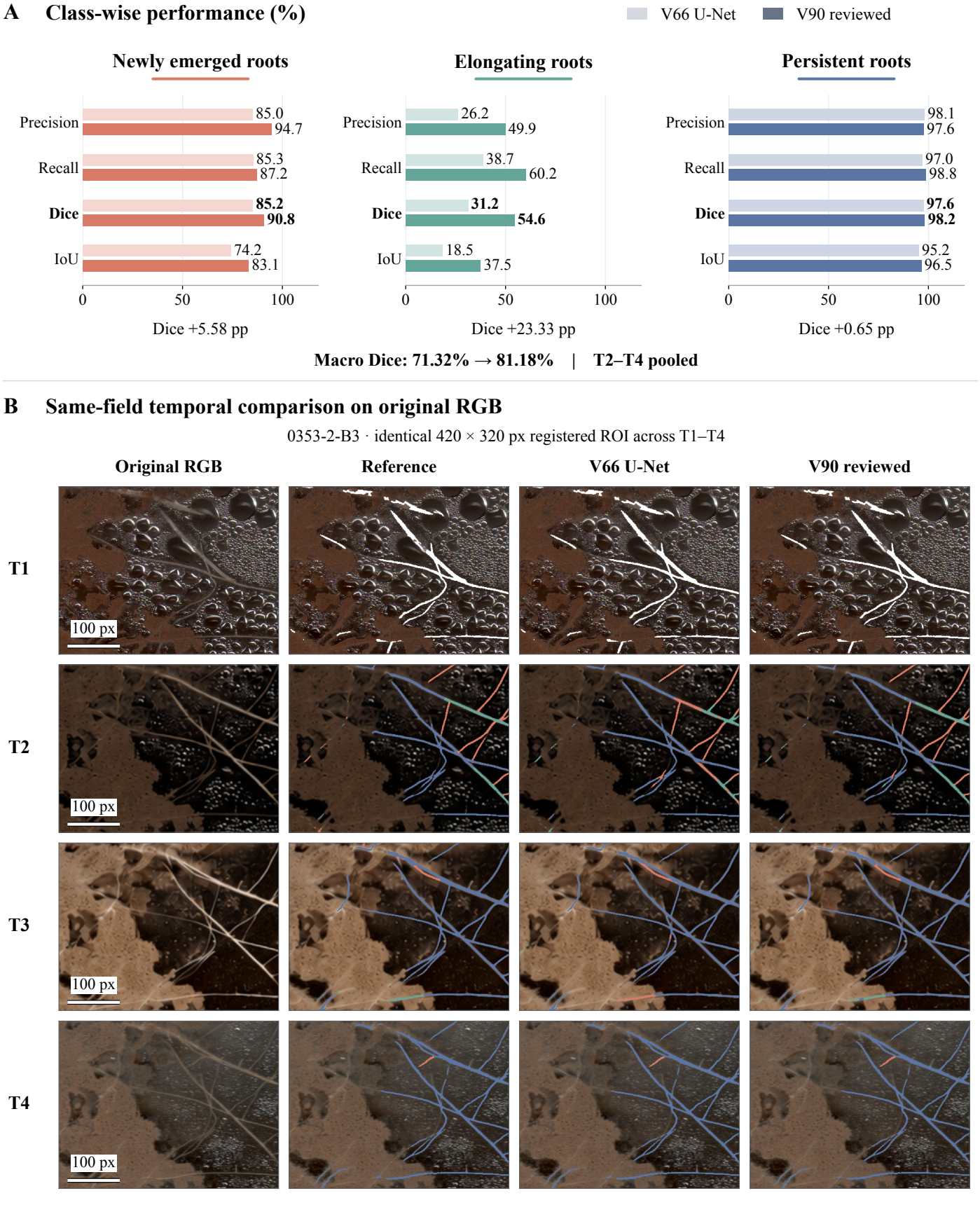

A
Class-wise performance (%)
V66 U-Net
V90 reviewed
Newly emerged roots
Elongating roots
Persistent roots
Precision
Recall
Dice
IoU
85.0
94.7
85.3
87.2
85.2
90.8
74.2
83.1
26.2
49.9
38.7
60.2
31.2
54.6
18.5
37.5
98.1
97.6
97.0
98.8
97.6
98.2
95.2
96.5
0
50
100
Dice +5.58 pp
Dice +23.33 pp
Dice +0.65 pp
Macro Dice: 71.32% → 81.18% | T2–T4 pooled
B
Same-field temporal comparison on original RGB
0353-2-B3 · identical 420 × 320 px registered ROI across T1–T4
Original RGB
Reference
V66 U-Net
V90 reviewed
T1
T2
T3
T4
100 px
Newly emerged
Elongating
Persistent
T1 baseline
Ignored reference
T1 is baseline; temporal-class metrics use T2–T4. Scale bars: 100 px.